Response to Reviewer 2

Manuscript ID 94388: FedRBC: Blockchain-Enabled Reputation-Weighted Robust Federated Learning for Privacy-Preserving Healthcare IoT Edge Networks

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Dear Reviewer 2,

We thank you for a demanding and precise review — the five major points and the additional requirements have reshaped this paper for the better. In response, we added the centralized and clean-FedAvg references that calibrate the operating regime; added FLTrust, Krum, and trimmed mean as first-class baselines and relabelled the simplified recent-system variants as aggregation-level re-implementations; extended the threat model with exactly the attacks that target this defence (LIE, backdoor scored by attack success rate, root-set perturbation, and the Byzantine ratio up to 0.5); measured the previously argued contributions (the ledger-memory ablation and the client-scaling study); restructured the theory around a new Lemma 1 that derives the residual malicious weight from the gate and ledger explicitly; fixed the kappa/beta symbol collision at its root; added an accounted differential-privacy mechanism backing the title's claim; and confined the device-feasibility language to emulation. We respond point by point below; all section/equation/figure/table numbers refer to the revised manuscript.

A note on the PDF rendering issues: several "misaligned tables" and "lost symbols in Eqs. 1, 6, 8, 9 and Theorem 1" reported by Reviewer 2 do not reproduce in our source PDF. Our LaTeX source compiles cleanly (0 errors, 0 undefined references), and we have verified the source characters one by one (e.g., Table 1 contains "identity", not "identiy"; Table 3's parenthesis is closed). We suspect a font/lossy-conversion artifact in the review-system's PDF rendering; we would be happy to supply the high-resolution source PDF and per-equation screenshots. The genuine typographical issues we did find (κ symbol collision, legend occlusion, shaded-region explanation) are all fixed in this revision regardless.

R2.1 — Results sit in a regime where the detector does not work; no centralized or clean-FedAvg upper bound; no-attack F1 below attacked F1; reconcile the ordering

Response. The upper bounds are now in the paper, and they reframe the regime correctly. Centralized training (identical model, lr 0.01, 30 epochs) reaches **F1 = 0.606 ± 0.062 at its final round (0.665 ± 0.025 at its best epoch)** on WUSTL-EHMS-2020 — FedRBC's attacked-regime 0.558–0.567 sits at **92–94% of the centralized final-round ceiling**. Clean FedAvg over the same partitions reaches only 0.401 ± 0.368 (it is itself unstable across seeds — see R1.9), which shows the difficulty is the task and the partitioning, not the defence. On the ordering puzzle (no-attack below attacked): the no-attack regime aggregates *all* skewed-but-benign non-IID updates at O(1) gate weight — this is the deliberate design choice that preserves honest information under heterogeneity, and its cost is a noisier quorum; under sign-flip, the gate suppresses precisely the direction-opposed mass *and* the reputation ledger removes those clients from selection in later rounds, so the effective quorum is cleaner and the *variance across seeds drops* (±0.05–0.07 attacked vs the no-attack spread). We report both regimes honestly with the new 10-seed numbers and discuss the interpretation in Section 5.7. A defence that keeps a cleaner quorum under attack than a heterogeneity-noisy one is not a contradiction — but we agree it demanded the upper bounds and per-seed transparency the revision now provides.

Changes. Table 4 (centralized + FedAvg upper bounds, 10 seeds); Section 5.7 (ordering analysis); per-seed disclosure.

R2.2 — FLTrust is missing although Eqs. (4)–(6) are FLTrust with a ReLU gate replaced by softmax; RAID/FedGT were reimplemented against FedRBC's own root reference; add FLTrust, Krum, or median/trimmed mean, or relabel

Response. All requested mechanisms are added as first-class baselines: **FLTrust** (with the same root set FedRBC uses — the comparison the reviewer designed — at 10 seeds), **Krum**, and **trimmed mean**, each under all three attack classes with 5–10 seeds. The relabelling is also done: RAID and FedGT entries are marked as aggregation-level re-implementations (ablated variants of the design space evaluated against FedRBC's root reference), per R1.7–R1.8. The softmax-vs-ReLU contrast the reviewer highlights is exactly what the new ablation and Section 4.3's discussion quantify: the softmax gate keeps skewed-but-benign non-IID updates that a hard ReLU truncates, at a bounded cost in the no-attack regime; FLTrust's ReLU gate is the natural canonical representative of the hard-gating family and is now measured alongside. The 10-seed comparison is as follows: FLTrust ends in a statistical tie with FedRBC on F1 (0.570/0.567/0.557 vs 0.567/0.561/0.558; Holm-corrected $p = 1.0$ in all three attacks) — we state the tie plainly rather than claim a mean advantage the seeds do not support — while FedRBC leads on AUC in every scenario (0.881/0.879/0.870 vs 0.857/0.857/0.862), on F1 variance in every scenario (0.049–0.066 vs 0.071–0.083), at the extended operating points (β=0.5: 0.519/0.564 vs 0.315/0.384 with collapsed seeds; N=50: 0.523 vs 0.375 with a collapsed seed), and is the only method in the comparison carrying trust memory, immutable auditability, and network-level containment.

Changes. Section 5.2 (baseline taxonomy); Tables 3–4 (FLTrust/Krum/trimmed mean); relabelled RAID/FedGT entries.

R2.3 — The threat model omits the attacks that target this defence; add one adaptive root-aware attack, one backdoor scored by ASR, and a root-set perturbation study

Response. All three are in the revision (new Section 5.9, Table 6): (i) the **LIE** attack (Baruch et al.) — collusive, per-coordinate mean-plus-z-sigma perturbation crafted to survive norm filters and stay direction-aligned with the honest mean, i.e., an adaptive attack aimed *at the gate's statistic* — breaks the plurality- and coordinate-based methods (EIFFeL 0.290±0.286, FedAvg 0.238±0.297, both with seed-level collapses) while the root-referenced family holds: FLTrust is strongest (0.605±0.066) and FedRBC stays at 0.517±0.054 — 96% of its own no-attack mean with the smallest variance of the four, a graceful degradation consistent with Lemma 1's prediction that shrinking Δc degrades the exponential factor rather than breaking the defence; (ii) a **targeted backdoor** with trigger injection scored by **attack success rate** alongside main-task F1: all methods preserve main-task F1 (FedRBC 0.604±0.099) but ASR is 0.994–0.996 across all methods — we report this honestly as a boundary of the design space (a planted trigger does not violate update direction, so no directional, magnitude, or reputation evidence discriminates it; suppression requires complementary anomaly-detection mechanisms outside this paper's scope, and the scope statement of Section 4.6 now says so); (iii) the **root-set perturbation study** ($\varepsilon \in 0.1$, 0.2, 0.3 on root-direction corruption under concurrent attacks, R1.2): FedRBC holds 0.566±0.101 F1 at ε=0.3 under Sybil — 95–104% of its clean-reference mean across the sweep — because the gate degrades but Layer-0, relative clipping, and the accumulated ledger continue to score and demote the adversary, whereas FLTrust, whose ReLU gate *solely* trusts the root direction, decays monotonically (0.522→0.467→0.417). The sweep beyond 40% Byzantine ratio is included ($\beta = 0.5$ for four methods × two attacks × 3 seeds) to test the plurality-independence claim at its breaking point: **FedRBC is the only method that stays above 0.5 F1** (0.519±0.098 sign-

flip, 0.564±0.038 Sybil), while FLTrust falls to 0.315±0.310 / 0.384±0.316 with collapsed seeds, EIFFeL to 0.148/0.138, and FedAvg to 0.074/0.148; the direction-preserving scaling attack stays in Section 5.7.

Changes. Section 5.9 (all three studies, Table 6); Section 5.4 (pointer to the β=0.5 breaking-point extension, reported in Section 5.9); Section 4.6 (scope).

R2.4 — Contributions C2 and C3 are argued, not measured: ledger ablation missing; Fig. 9 is M/M/1 emulation with author-chosen cloud parameters; component ablation grid and client-scaling study needed; gas saturation near 150 clients

Response. C2 is now measured: the **cross-round-memory ablation** (per-round reputation reset, isolating the persistent-ledger contribution from the per-round mechanism) is in the new ablation grid (Table 8), alongside the independent removal of gate/clip/Layer-0/reputation-weighting/selection. The honest result: resetting the ledger every round leaves final F1 statistically unchanged — which sharpens the ledger's case rather than weakening it, because the F1 defence is carried by the gate while the ledger's measured contribution is exactly the operational dimension: the round-60 reputation separation (honest median 0.109 vs Byzantine 0.009, a 12× gap that keeps the selection ranking intact), the impossibility of consequence-resetting for intermittent adversaries, the immutable audit trail, and the containment events. C3's measurement honesty is fixed as described in R1.12 — Fig. 9 is labelled an emulation, the cloud parameters' provenance is stated, and the abstract's 50–80% claim is qualified. The **client-scaling study** (N = 20, 50, 100) is added: FedRBC's variance *shrinks* with population (0.073 at N=50 → 0.037 at N=100, F1 0.523 → 0.573) because the density-greedy selection keeps the aggregation quorum at K regardless of N, while FLTrust shows a collapsed seed at N=50 (0.375±0.310) and clean FedAvg degenerates at both scales; the gas-saturation boundary the reviewer computed (≈150 clients/round at 48.6k gas/write) is now stated explicitly in Section 5.6 and connected to the scaling study.

Changes. Ablation grid (ledger memory); Fig. 9 caption; Section 5.6 (gas boundary); client-scaling study.

R2.5 — Theorem 1 assumes the residual malicious weight Eq. (6) should bound (circular); Prop. 1/2 and Thm. 2 restate standard results; β=0.6 vs 2.0 contradiction; seed counts differ; std = 0.000; FedGT collapsed row; no multiple-comparison correction

Response. Each item, in order:

1. *Circularity — fixed at the root.* The reviewer is right, and this was the revision's main theoretical change. Lemma 1 (Eq. 15) now **derives** the residual malicious weight from the gate and ledger: $\epsilon_r(t) \leq \rho_\beta \rho_p \rho_r e^{-\Delta c/T_s}$, an explicit product of the Byzantine ratio, the prior ratio, the reputation ratio, and the directional-separation gap — none of it assumed. Theorem 1 consumes this bound; Appendix B.1 proves both. The ledger factor contracts geometrically by Prop. 1, which is what makes the adversarial term *tighten* as reputation learning proceeds — the previous version gestured at this; the new version proves it.

2. *Standard results.* We agree Prop. 1's contraction, Prop. 2's greedy bound, and Thm. 2's M/M/1 sojourn are elementary in isolation, and the revision says so where relevant; the contribution claimed for each is the mechanism-specific parameterization (the penalty-branch product $\kappa\gamma$ and the intermittent-penalty doubling analysis; the density rule under reputation-latency constraints; the edge/cloud architectural separation), and each is now *empirically calibrated* (measured T0, measured overheads, emulated MTTC with stated parameters). The genuinely new theoretical object in the paper is Lemma 1.

3. *β = 0.6 vs 2.0.* This was a symbol collision, now fixed globally: the clipping coefficient is $\kappa_c = 2$ (Eq. 5)

and the penalty-acceleration product is $\kappa\gamma = 0.6$ (Eq. 9) — deliberately distinct symbols, declared at first use, audited in every occurrence (Eq. 5, Alg. 1, Table 2, Thm. 1, Appendix B.1).

4. *Seed counts / std = 0.000 / collapsed rows.* All headline results are 10-seed with 95% CI; the exactly-zero stds came from collapsed seeds' degenerate single-class predictions — root-caused (NaN propagation; R1.9), fixed in code, and the previously collapsed seeds are reported transparently (including FedGT's Sybil row: F1 collapse with majority-class accuracy is the signature of its hard filter discarding all updates in those seeds, which the per-seed table makes visible).

5. *Multiple testing.* Holm's step-down correction is applied across the nine pairwise comparisons; corrected p-values are in the main table.

Changes. Section 4.6 (Lemma 1, restructured Thm. 1, scope); Appendix B.1; global κ_c audit; 10 seeds + CI; Holm correction; per-seed disclosure.

R2 "Also required" — The title claims "Privacy-Preserving" but no privacy mechanism exists

Response. We keep the claim and now back it with an accounted mechanism. The revision adds a Gaussian output-perturbation layer on the aggregated step with an explicit RDP-composed (ε, δ) budget per training job (new Section 4.7, Eqs. 16–17: the per-round Rényi loss is $\alpha/(2\sigma_{\mathrm{dp}}^2 K)$ under uniform weights, with a conservative $1/q_{\max}$ calibration for the gated weights), and measures the privacy–robustness interaction across three noise levels × four methods × two attacks (Section 5.9, Table 7). The headline trade-off: at $\sigma_{\mathrm{dp}} = 0.005$–$0.02$ the layer costs FedRBC at most a few points (0.608/0.549 and 0.585/0.510 F1 under sign-flip/Sybil — essentially its no-DP means, with the lightest level acting as a mild regularizer against sign-flip), and at the stress level $\sigma_{\mathrm{dp}} = 0.1$ FedRBC retains 0.447–0.448 while FLTrust's hard gate collapses to 0.220–0.222; the measured curve locates deployment working points, and Eq. (17) turns each σ_{dp} into an explicit budget. The mechanism's scope is stated honestly: it protects the *aggregated model trajectory* — the gradient-inversion surface; per-update confidentiality against an honest-but-curious aggregation server additionally requires secure aggregation, which we discuss as a composed-future-work item in Section 6. The remaining exposure boundaries are by design: the root set is never trained on and never leaves the server, and the ledger stores 5-byte scalar evidence only (no gradients, no identifiers — Section 4.1 states the ledger's privacy envelope explicitly).

Changes. Section 4 (DP mechanism + budget); Section 5.9 (DP experiments); Section 4.1/6 (exposure discussion).

R2 "Also required" — Tables 1–4 misaligned in the PDF; Eqs. 1/6/8/9 and Theorem 1 lost symbols

Response. These artifacts do not reproduce from our source: the LaTeX compiles with 0 errors/0 undefined references, and character-level inspection of the source shows the reported truncations absent ("identity" spelled in full; parentheses closed; all symbols present in Eqs. 1/6/8/9 and Thm. 1). We believe the review system's PDF rendering (font substitution/lossy conversion) caused the artifacts. We supply the high-resolution source PDF with this resubmission and per-equation/per-table screenshots in the response package; if the editorial office indicates a specific page where the misalignment appears in their rendered copy, we will adapt the source to survive that pipeline.

R2 "Also required" — Twenty logical clients on CUDA streams support no device-feasibility claim; confine language to emulation

Response. Done. All feasibility language is now confined to the emulated testbed: Section 5.1 states that the evaluation emulates N logical clients (with the client-scaling study making the emulation parameter explicit), and every "deployment" claim is either scoped to the emulation or moved to future work with the concrete missing measurements named (Section 6).

R2 "Literature to engage with" — Cite the two fielded systems

Response. Both are now cited and engaged where they are most on point: [33] (blockchain-secured IoT-federated learning for industrial air-pollution monitoring, *Ecotoxicology and Environmental Safety*, 2025) and [34] (federated IoT framework for indoor air quality / HVAC optimization in healthcare buildings, *Journal of Building Engineering*, 2025) are introduced in Section 1.2 as the first fielded blockchain-FL and building-scale FL deployments, and Section 6.1 uses them as the fielded-deployment reference point for the operational claims (gas cost, write cadence, containment actuation) that previously rested on the literature alone. DOIs verified via CrossRef.

Changes. Section 1.2 (deployment lineage); Section 6.1 (fielded comparison); References (two new entries, verified).